

Yingrui Hou
Clermont-Ferrand, France
+33 (0)7 85 39 63 32
yingrui.hou@outlook.com
<https://yingrui-hou.github.io/PersonalPage/>

24 August 2026

Hiring Team
Limagrain Vegetable Seeds Traits and Technology
Chappes, France

Subject: Application for Biostatistician - AI & Data Science for Breeding Innovation

Dear Hiring Team,

I am applying for the Biostatistician - AI & Data Science for Breeding Innovation position in Chappes. I hold a PhD in experimental physics and have several years of postdoctoral experience developing reusable Python/C++ workflows for statistical modelling, machine-learning validation, time-dependent inference, and decision support on large, noisy datasets. The opportunity to transfer these methods to genomic, phenotypic, and environmental data, while working directly with breeders, geneticists, and data teams, strongly matches the direction I want to take in industry.

My research has repeatedly required reliable conclusions from incomplete or biased observations. I implemented probabilistic and maximum-likelihood models, developed CatBoost pipelines with train/test diagnostics, aligned mismatched datasets through gradient-boosting reweighting and control-sample calibration, and contributed covariance-aware checks, pseudoexperiments, and robustness studies. I also built monitoring workflows that converted model discrepancies into calibration follow-up and software corrections. This work required not only statistical depth but also reproducible implementation in Linux-based shared codebases and clear communication across large international collaborations.

I am particularly attracted by the role's combination of methodological research and deployable decision-support tools. My current strongest data-science language is Python rather than R, and my domain background is experimental physics rather than quantitative genetics. I would treat advanced R, breeding methodology, plant-science data, and LLM-based tools as explicit development priorities. The underlying analytical problems are already familiar: heterogeneous datasets, latent variables, limited ground truth, distribution shift, uncertainty, and the need to make predictions trustworthy for expert decisions.

My transition from academia is deliberate: I am now clear that my main motivation lies in solving complex real-world problems, not in particle physics itself. I can contribute immediately to statistical modelling, ML validation, reproducible analytical modules, and model credibility while developing the domain and deployment capabilities required by the team. Applying quantitative methods to more efficient and sustainable breeding gives that transition a concrete scientific purpose.

I would welcome the opportunity to discuss how my experience in statistical inference, validation under dataset shift, and collaborative scientific software could contribute to Limagrain's next generation of breeding analytics.

Yours sincerely,

Yingrui Hou